



SteadySense

An Intelligent Real-Time Elder Care Application

User Manual

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Contents

1. Getting Started
2. Account Setup
3. Core Features
4. Emergency Features
5. Caretaker Dashboard
6. Voice Assistant
7. Settings and Permissions
8. Troubleshooting
9. Support

1 Getting Started

1.1 Installation

1. Download SteadySense from the Google Play Store or APKPure
2. Open the application and allow necessary permissions when prompted
3. You will be directed to either register as a new user or login

1.2 System Requirements

- Android 8.0 or higher
- Minimum 100 MB free storage space
- Active internet connection (WiFi or mobile data)
- GPS enabled for location tracking features

2 Account Setup

2.1 Registration for Elders

1. Tap **Register** on the login screen
2. Enter your full name and email address
3. Create a secure password (minimum 6 characters recommended)
4. Verify your email address through the confirmation link
5. Select **I am an Elder** during role selection
6. Your account is now ready to use

2.2 Registration for Caretakers

1. Tap **Register** on the login screen

2. Enter your full name and email address
3. Create a secure password
4. Verify your email address
5. Select **I am a Caretaker** during role selection
6. You will receive a 6-digit Caretaker ID to share with elders

2.3 Linking Caretakers to Elders

1. Open SteadySense as an Elder account
2. Navigate to **Settings** → **Linked Caretakers**
3. Tap **Add Caretaker**
4. Enter the 6-digit ID provided by your caretaker
5. Confirm the connection
6. Your caretaker now has access to your health data and emergency alerts

3 Core Features

3.1 Health Dashboard

The Health Dashboard displays real-time information about the elder's well-being:

- **Daily Step Count:** Tracks the number of steps taken throughout the day
- **Activity Status:** Shows current activity (Resting, Walking, or Inactive)
- **Heart Rate:** Displays simulated pulse information for health monitoring
- **Time Since Last Activity:** Indicates how long the device has been stationary

3.2 Real-Time Location

- GPS coordinates are continuously tracked when the app is active
- Location updates occur every 30 seconds during normal operation
- High-accuracy location is shared with caretakers during emergency events
- MapLibre GL provides fast, interactive map visualization

3.3 Activity Monitoring

- Device accelerometer continuously monitors movement patterns
- Step count accumulates throughout the day (resets at midnight)
- Activity transitions are automatically detected and recorded

4 Emergency Features

4.1 Automatic Fall Detection

SteadySense continuously monitors for falls using the device's built-in accelerometer:

- Falls are detected by analyzing sudden impacts followed by inactivity
- When a potential fall is detected, a 30-second countdown begins
- During this time, the elder can dismiss the alert by tapping **I'm OK**
- If not dismissed, all linked caretakers receive an emergency notification with the elder's location
- The app vibrates and plays an alert sound to notify the elder

4.2 Manual Emergency Alert (SOS)

1. Press and hold the large red **SOS** button on the main screen
2. Maintain pressure for 3 seconds to activate
3. All linked caretakers will immediately receive an emergency alert
4. Your precise GPS location is shared on their map dashboard

5. The app will continue sharing location updates until you clear the alert

4.3 Responding to Alerts

Elders have 30 seconds to respond after a fall is detected:

- Tap **I'm OK** if you are safe and no help is needed
- If no response is given, emergency alerts are automatically sent
- You can also tap **Need Help** to confirm you require assistance

5 Caretaker Dashboard

5.1 Monitoring Linked Elders

Caretakers can monitor multiple elders from a single dashboard:

- List of all linked elders with current status (Online, Last Seen)
- Real-time step count and activity status for each elder
- Quick access to view an elder's current location on the map
- Emergency alert history and response logs

5.2 Live Map View

- View an elder's location on an interactive map
- Location updates refresh every 30 seconds
- Map includes street names, landmarks, and terrain
- Zoom and pan to adjust view as needed

5.3 Receiving Emergency Alerts

When an emergency occurs:

1. You receive a high-priority push notification
2. The elder's name and type of emergency (Fall or SOS) is displayed
3. Their precise GPS location is automatically opened on a map
4. You can contact the elder or emergency services immediately

5.4 Unlinking Elders

1. Open the caretaker dashboard
2. Select an elder from the list
3. Tap **Settings** → **Unlink Elder**
4. Confirm the action (this will remove access to that elder's data)

6 Voice Assistant

6.1 Accessing the Voice Assistant

1. Tap the microphone icon on the main screen
2. Grant microphone permissions if prompted
3. Wait for the listening indicator to appear
4. Speak your query or request clearly

6.2 Supported Languages

- English: Full support for conversational queries
- Urdu: Full support for conversational queries
- Switch languages in **Settings** → **Language Preferences**

6.3 Sample Voice Commands

- “What’s my step count today?”
- “Is it going to rain?”
- “Tell me a joke”
- “Set a reminder for 2 PM”
- “Call my caretaker”

6.4 Voice History

1. Open the app and tap **Voice History**
2. View a log of all previous voice interactions
3. Audio is not stored—only conversation summaries are recorded
4. You can clear history anytime in **Settings**

7 Settings and Permissions

7.1 Accessing Settings

1. Tap the gear icon in the top-right corner
2. Navigate through available options

7.2 Required Permissions

SteadySense requires the following permissions to function properly:

- **Location (Foreground & Background):** Essential for sharing your location during emergencies and activity tracking
- **Sensor Access:** Required to detect falls and count steps
- **Microphone:** Necessary for voice assistant interactions
- **Notifications:** Required to receive emergency alerts and system notifications
- **Contacts:** Optional—for quick emergency contact calling

7.3 Permission Management

1. Go to **Settings** on your Android device
2. Find **Applications** → **SteadySense**
3. Tap **Permissions**
4. Enable or disable permissions as needed (not recommended to disable critical permissions)

7.4 Notification Settings

- Tap **Settings** → **Notifications**
- Toggle notification types: Emergency Alerts, Activity Reminders, System Updates
- Adjust vibration and sound preferences

7.5 Background Service Management

- Enable **Foreground Service** to ensure fall detection runs continuously
- Disable battery optimization for SteadySense to prevent interruptions
- Note: Disabling the background service will prevent fall detection

8 Troubleshooting

8.1 App Crashes or Won’t Start

1. Force close the app: **Settings** → **Apps** → **SteadySense** → **Force Stop**

2. Clear app cache: **Settings** → **Apps** → **SteadySense** → **Storage** → **Clear Cache**
3. Restart your device
4. Reinstall the app if the problem persists

8.2 Location Not Updating

1. Ensure GPS is enabled on your device
2. Confirm location permissions are granted to SteadySense
3. Restart the app and wait 1-2 minutes for the first update
4. Check that you have an active internet connection
5. Move to an open area away from tall buildings if GPS signal is weak

8.3 Fall Detection Not Working

1. Verify the app has sensor access permissions
2. Ensure the app is not force-stopped by Android battery optimization
3. Check that foreground service is enabled in app settings
4. Restart your device
5. Note: Extreme impacts may take a few seconds to register

8.4 Not Receiving Emergency Alerts (Caretaker)

1. Confirm notifications are enabled in app settings
2. Check device notification settings for SteadySense
3. Verify you are properly linked to the elder (check 6-digit connection)
4. Ensure you have an active internet connection
5. Restart the app and device if alerts are still not arriving

8.5 Voice Assistant Not Responding

1. Grant microphone permissions if not yet granted
2. Verify your internet connection (voice processing requires connectivity)
3. Speak clearly and wait for the listening indicator to appear
4. Check that the selected language matches your speech
5. Restart the app if voice assistant becomes unresponsive

8.6 Can't Link Caretaker

1. Verify the 6-digit caretaker ID is correct
2. Ensure both accounts are registered and email-verified
3. Check that the caretaker account exists and is active
4. Try again after 5 minutes if the system shows a temporary error
5. Contact support if the issue persists

9 Support

9.1 Frequently Asked Questions

Q: How often is my location updated?

A: Location updates occur every 30 seconds during normal operation and continuously during emergency events.

Q: Is my data secure?

A: Yes. All data is encrypted and stored on Firebase, a secure Google infrastructure. Only linked caretakers have access to your information.

Q: Can I have multiple caretakers?

A: Yes. You can link multiple caretakers, and all will receive emergency alerts simultaneously.

Q: Does SteadySense work offline?

A: Core features like fall detection work offline, but emergency alerts and location sharing require an active internet connection.

Q: What happens to my data if I delete my account?

A: All personal data is permanently deleted from our servers. This action cannot be undone.

9.2 Contact Information

For additional support, technical issues, or feature requests, please contact us through the app's **Help** section or visit our website.

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